

Data Structure and Algorithm - 2

Technical Workouts

1. Understand Basic Sorting Algorithms & Hash Tables

- a. What is sorting
- b. Bubble sort, Selection sort and Insertion sort
- c. Merge sort and Quick sort
- d. Time and space complexity of sorting algorithms
- e. One-page revision table (choosing the right sort)

2. Understand Hash Table

- a. What is a hash table
- b. Hashing and hash function
- c. Collision
- d. Separate chaining (open hashing)
- e. Open addressing: linear probing, quadratic probing, double hashing

3. Understand Stacks & Queues

- a. Stack and LIFO principle
- b. Stack operations: push, pop and display
- c. Queue and FIFO principle
- d. Queue operations: enqueue, dequeue and display
- e. Applications of stack and queue

4. Understand Trees & Binary Search Trees

- a. Tree concepts
- b. Binary tree
- c. Binary search tree (BST)
- d. BST operations: insertion, contains, delete and traversals
- e. Closest value in a BST and validating a BST

5. Understand Graphs & Traversals

- a. Graph concepts
- b. Types of graphs: undirected, directed and weighted
- c. Applications of graphs
- d. BFS (Breadth First Search)
- e. DFS (Depth First Search)